

Hands-on Lab: Database Design using ERDs



Estimated time needed: 45 minutes

In this lab, you will learn how to design a database by creating an entity relationship diagram (ERD) in the PostgreSQL database service using the pgAdmin graphical user interface (GUI) tool. First, you will create an ERD of a database. Next, you will generate and execute an SQL script to create the database schema from its ERD. Finally, you will load the created database schema with data.

Software used in this lab

In this lab, you will use [PostgreSQL Database](#). PostgreSQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



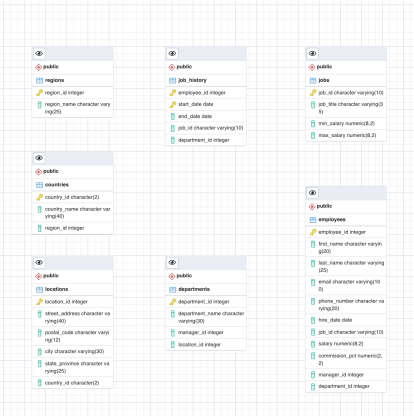
To complete this lab, you will utilize the PostgreSQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Database used in this lab

The HR database used in this lab comes from the following source: [HR Sample Database](#) [Copyright 2021 - Oracle Corporation].

You will use a modified version of the database for the lab. To follow the lab instructions successfully, please use the database provided with the lab, rather than the database from the original source.

The following ERD shows the tables of the HR database:



Objectives

After completing this lab, you will be able to use pgAdmin with PostgreSQL to:

- Create an ERD of a database.
- Generate and execute an SQL script from an ERD to create a schema.
- Load the database schema with data.

This lab is divided into two exercises, *Example Exercise* and *Practice Exercise*.

Example Exercise

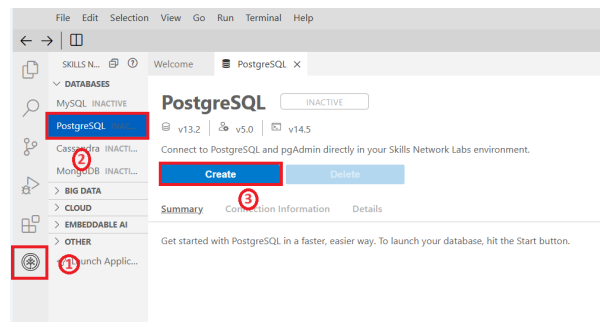
In this example exercise, you will first create a partial ERD of the HR database. Next, you will generate and execute an SQL script to create the partial schema of the HR database from its ERD. Finally, you will load the created database schema with data by using the Restore feature.

Task A: Create an Entity Relationship Diagram (ERD) of a database

In this task of the Example Exercise, you will create a partial ERD of the HR database.

To get started with this lab, launch PostgreSQL using the Cloud IDE. You can do this by following these steps:

1. Click the Skills Network extension button on the left side of the window.
2. Open the **DATABASES** menu and click **PostgreSQL**.
3. Click **Create**. PostgreSQL may take a few moments to start.



4. Note down your PostgreSQL service session password because you may need to use it later in the lab.

5. Click the pgAdmin button in the same window where you started PostgreSQL.

6. You will see the pgAdmin GUI tool.

pgAdmin

File ▾ Object ▾ Tools ▾ Help ▾

Browser

Servers

Dashboard

Properties

SQL

Statistics

Dependencies

Dependents

Welcome

 **pgAdmin**
Management Tools for PostgreSQL

Feature rich | Maximises PostgreSQL | Open Source

pgAdmin is an Open Source administration and management tool for the PostgreSQL database. It includes a graphical administration interface, an SQL query tool, a procedural code debugger and much more. The tool is designed to answer the needs of developers, DBAs and system administrators alike.

Quick Links


Add New Server


Configure pgAdmin

Getting Started


PostgreSQL Documentation


pgAdmin Website


Planet PostgreSQL


Community Support

7. In the tree-view, expand **Servers** > **postgres** > **Databases**. Enter your PostgreSQL service session password if prompted during the process. Right-click on **Databases** and go to **Create** > **Database**. Type **HR** as the name of the database and click **Save**.

The screenshot shows the pgAdmin 4 interface. In the left-hand tree view, the 'Servers (1)' folder is expanded, showing the 'postgres' server. The 'Databases (1)' folder under 'postgres' is also expanded, and the 'Create' context menu is open, with 'Database...' selected. The main pane displays the 'Server sessions' table with 7 rows and the 'Tuples in' table with 1 row. The 'Create - Database' dialog box is open, showing the 'General' tab. The 'Database' field is set to 'HR', the 'Owner' is 'postgres', and the 'Comment' field is empty. The 'Save' button is highlighted.

pgAdmin File ▾ Object ▾ Tools ▾ Help ▾

Browser 1 Servers (1) 2 postgres 3 Databases (1) postgres Create Database... Cast Catalog Event Triggers Extensions Foreign Data Wrappers Languages Publications Schemas Subscriptions Login/Group Roles Tablespaces

Server sessions

7
4
3
2
1
0

Tuples in Inse

1

Create - Database

General Definition Security Parameters Advanced SQL

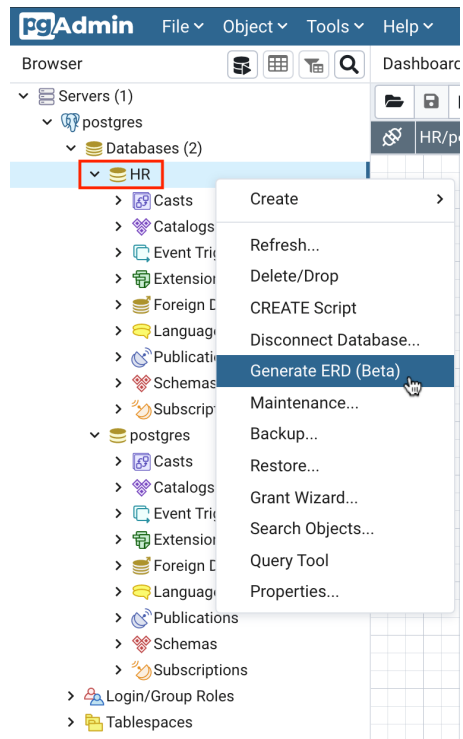
Database HR

Owner postgres

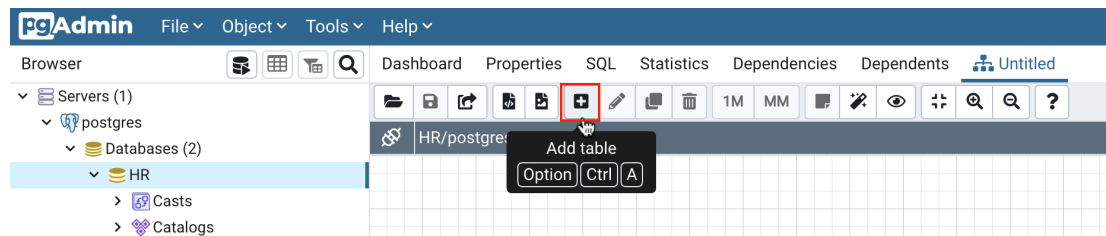
Comment

Cancel Reset Save

8. In the tree-view, expand **HR**. Right-click on **HR** and select **Generate ERD (Beta)**.



9. Click **Add table**. On the **General** tab, in the **Name** box, type **employees** as the name of the table. Don't click **OK**; proceed to the next step.



New table

General

Columns

Name

employees

Schema

public

Comment

✕ Cancel

OK

10. Switch to the **Columns** tab and click **Add new row** to add the necessary column placeholders. Now enter the **employees** table definition information as shown in the image below to create its entity diagram. Then click **OK**.

New table

General

Columns

Columns

Name	Data type	Length/Precision	Scale	Not NULL?	Primary key?
------	-----------	------------------	-------	-----------	--------------

✕ Cancel

OK

New table

General

Columns

Columns

	Name	Data type	Length/Precision	Scale	Not NULL?	Primary key?
 	employee_id	integer			Yes	Yes
 	first_name	character varying	20		No	No
 	last_name	character varying	25		Yes	No
 	email	character varying	100		Yes	No
 	phone_number	character varying	20		No	No
 	hire_date	date			Yes	No
 	job_id	character varying	10		Yes	No
 	salary	numeric	8	2	Yes	No
 	commission_pct	numeric	2	2	No	No
 	manager_id	integer			No	No
 	department_id	integer			No	No

✖ Cancel

OK

11. Similarly, create entity diagrams for the other three tables following steps 9 and 10:

► [Click here] Create an entity diagram for the jobs table

Click **Add table** icon. On the **General** tab, in the **Name** box, type **jobs** as the name of the table. Don't click **OK**. Switch to tab **Columns** and click **Add new row** to add the necessary column placeholders. Now enter the **jobs** table definition information as shown in the image below to create its entity diagram. Then click **OK**.

Table: jobs (public)

General

Columns

Name

jobs

Schema

 public

Comment

✖ Cancel

OK

Table: jobs (public)

General

Columns

Columns

Name

Data type

Length/Precision

Scale

Not NULL?

Primary key?

job_id

character varying

10

Yes

Yes

job_title

character varying

35

Yes

No

min_salary

numeric

8

2

No

No

max_salary

numeric

8

2

No

No

Cancel

OK

• [Click here] Create an entity diagram for the departments table

Click **Add table** icon. On the **General** tab, in the **Name** box, type **departments** as the name of the table. Don't click **OK**. Switch to tab **Columns** and click **Add new row** to add the necessary column placeholders. Now enter the **departments** table definition information as shown in the image below to create its entity diagram. Then click **OK**.

Table: departments (public)

General

Columns

Name

departments

Schema

public

Comment

Cancel

OK

Click **Add table** icon. On the **General** tab, in the **Name box**, type **locations** as the name of the table. Don't click **OK**. Switch to tab **Columns** and click **Add new row** to add the necessary column placeholders. Now enter the **locations** table definition information as shown in the image below to create its entity diagram. Then click **OK**.

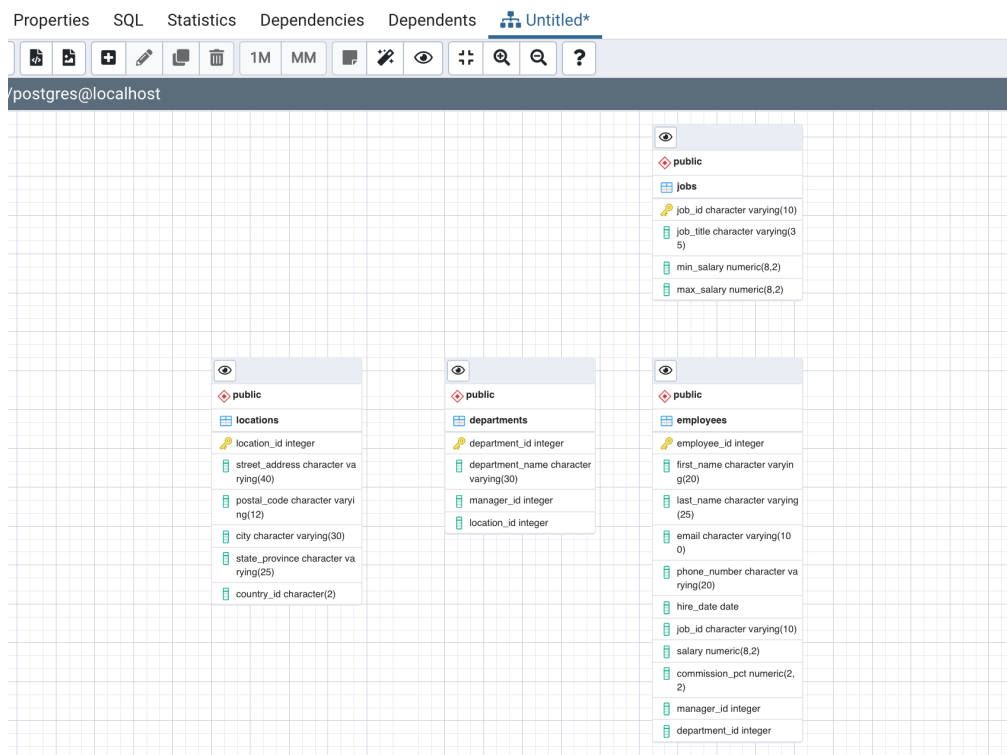
Table: locations (public)

General Columns

Columns

	Name	Data type	Length/Precision	Scale	Not NULL?	Primary key?
 	location_id	integer			☒ Yes	☒ Yes
 	street_address	character varying	40		☐ No	☐ No
 	postal_code	character varying	12		☐ No	☐ No
 	city	character varying	30		☒ Yes	☐ No
 	state_province	character varying	25		☐ No	☐ No
 	country_id	character	2		☒ Yes	☐ No

12. After creating all four entity diagrams, the entities of the ERD are complete.



13. Next, you will create relationships between the entities by adding foreign keys to the tables. Select the entity diagram **employees** and click **One-to-Many link**. Now enter the definition information for a foreign key on the **employees** table as shown in the image below to create the relationship. Then click **OK**.

The screenshot shows a database management interface with a schema diagram. The 'employees' table is highlighted with a red box. A 'One-to-Many link' tooltip is visible above the 'employees' table.

employees

- employee_id integer
- first_name character varying(20)
- last_name character varying(25)
- email character varying(100)
- phone_number character varying(20)
- hire_date date
- job_id character varying(10)
- salary numeric(8,2)
- commission_pct numeric(2,2)
- manager_id integer
- department_id integer

jobs

- job_id character varying(10)
- job_title character varying(35)
- min_salary numeric(8,2)
- max_salary numeric(8,2)

departments

- department_id integer
- department_name character varying(30)
- manager_id integer
- location_id integer

locations

- location_id integer
- street_address character varying(40)
- postal_code character varying(12)
- city character varying(30)
- state_province character varying(25)
- country_id character(2)

One to many relation

General

Local Table

(public) employees

Local Column

department_id

x

Referenced Table

(public) departments

x

Referenced Column

department_id

x

Cancel

OK

12. Similarly, create the other relationships between the tables following the instructions in step 13:

► [Click here] Create a relationship between employees and jobs

Select the entity diagram **employees** and click **One-to-Many link**. Now enter the definition information for a foreign key on the **employees** table as shown in the image below to create the relationship. Then click **OK**.

One to many relation

General

Local Table

(public) employees

Local Column

job_id

x

Referenced Table

(public) jobs

x

Referenced Column

job_id

x

Cancel

OK

► [Click here] Create a relationship between departments and locations

Select the entity diagram **departments** and click **One-to-Many link**. Now enter the definition information for a foreign key on the **departments** table as shown in the image below to create the relationship. Then click **OK**.

One to many relation

General

Local Table

(public) departments

Local Column

location_id

Referenced Table

(public) locations

Referenced Column

location_id

Cancel

OK

► [Click here] Create a relationship between departments and employees

Select the entity diagram **departments** and click **One-to-Many link**. Now enter the definition information for a foreign key on the **departments** table as shown in the image below to create the relationship. Then click **OK**.

One to many relation

General

Local Table

(public) departments

Local Column

manager_id

Referenced Table

(public) employees

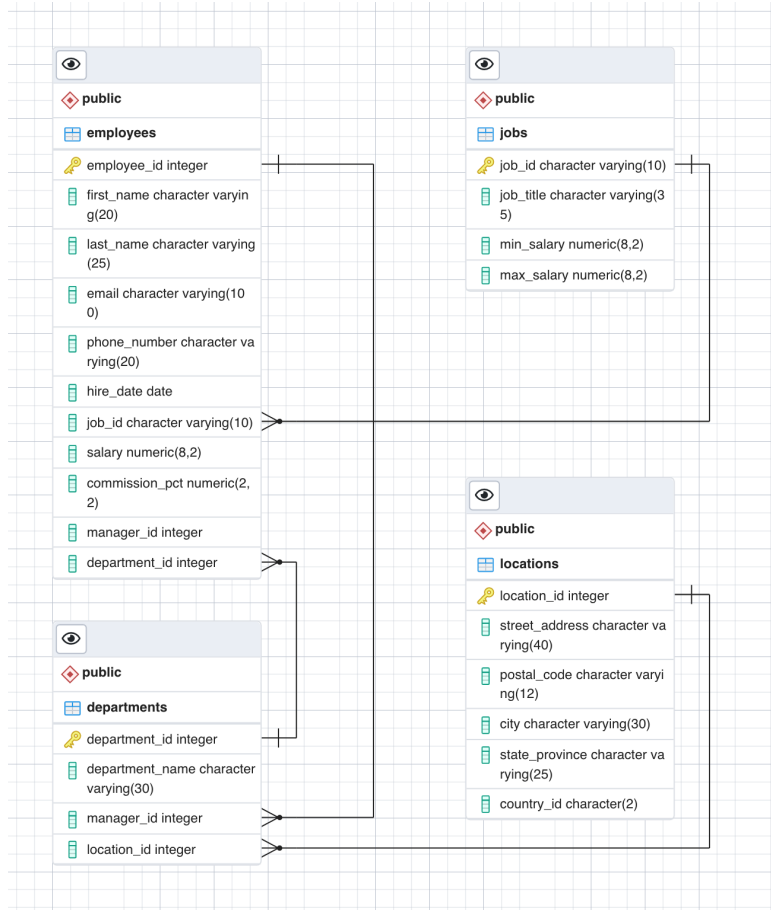
Referenced Column

employee_id

Cancel

OK

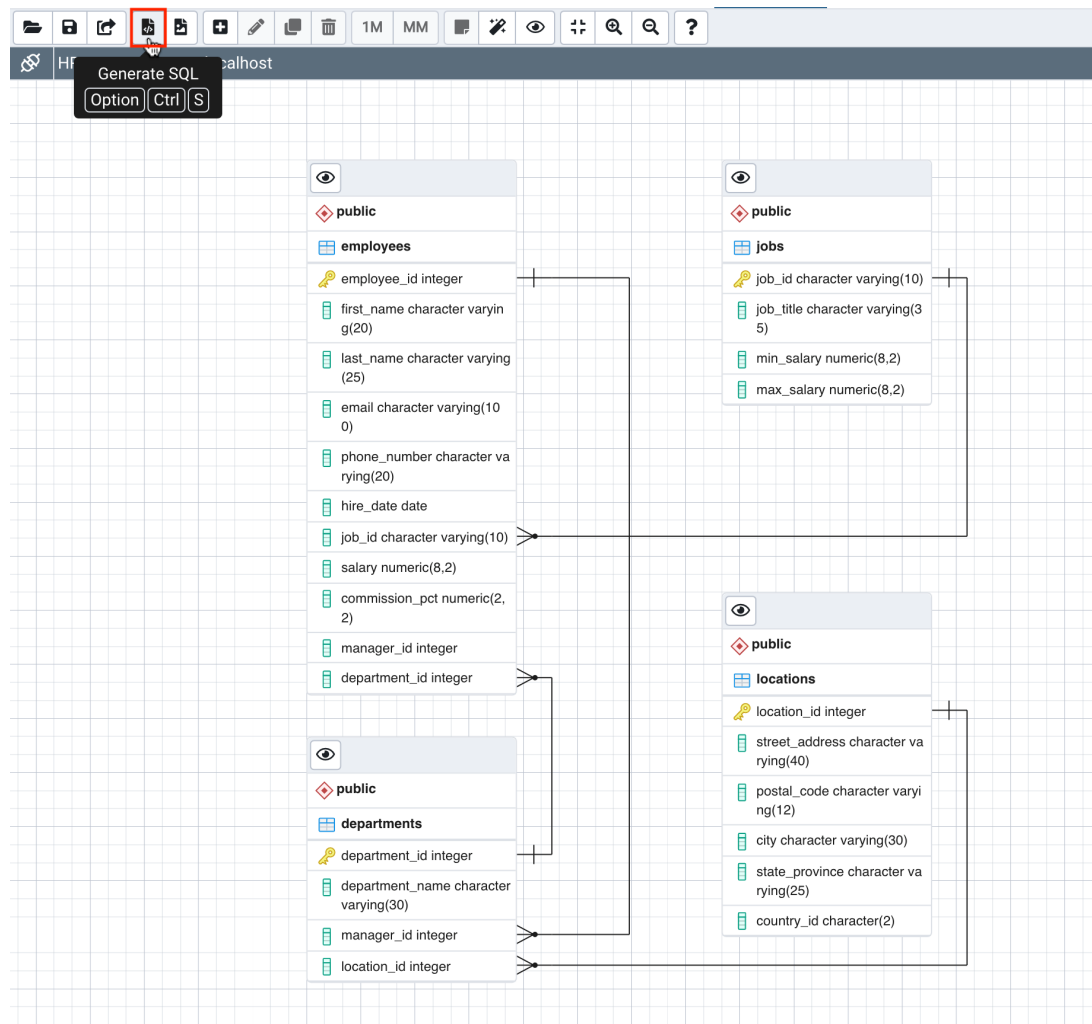
13. After creating all four relationships, you have completed the ERD for this exercise. Proceed to Task B.



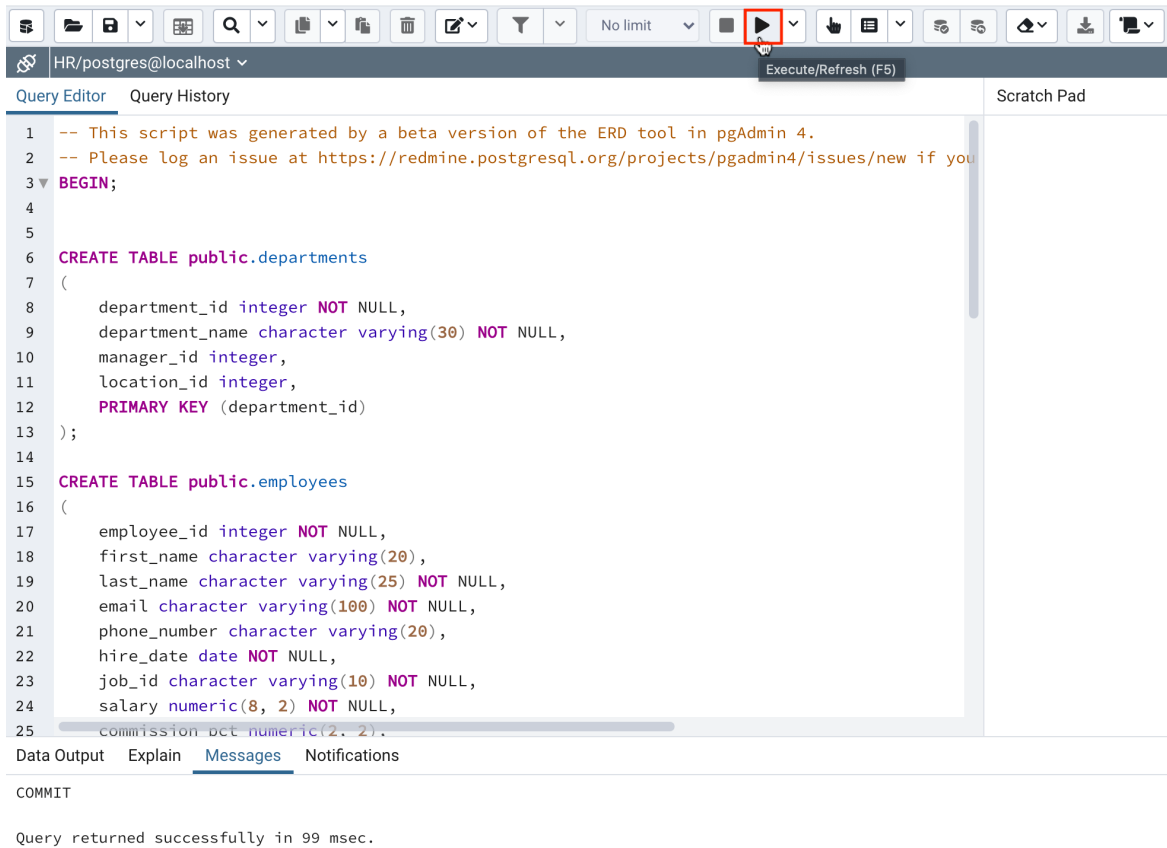
Task B: Generate and execute SQL script from ERD to create the schema

In this task of the Example Exercise, you will generate and execute a SQL script from the ERD you created in Task A of the Example Exercise.

- 1. In the **Generate ERD (Beta)** window, click **Generate SQL**.



2. A new Query Editor window will open containing a SQL script generated from the ERD. Click **Execute/Refresh** to run the script. Proceed to Task C.



The screenshot shows the pgAdmin 4 Query Editor interface. The top toolbar includes icons for file operations, search, and execution. The 'Execute/Refresh (F5)' button is highlighted with a red box. The main area displays a SQL script for creating two tables: 'public.departments' and 'public.employees'. The script is as follows:

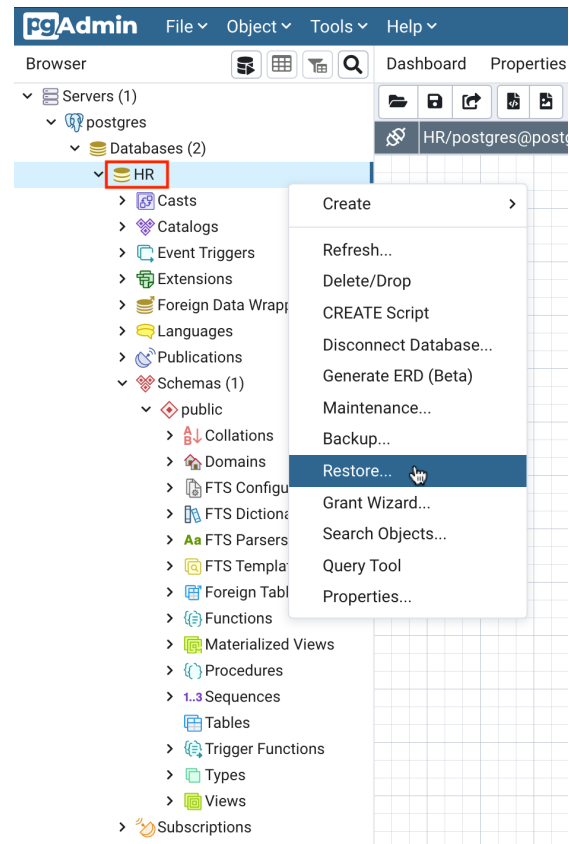
```
1  -- This script was generated by a beta version of the ERD tool in pgAdmin 4.
2  -- Please log an issue at https://redmine.postgresql.org/projects/pgadmin4/issues/new if you
3  BEGIN;
4
5
6  CREATE TABLE public.departments
7  (
8      department_id integer NOT NULL,
9      department_name character varying(30) NOT NULL,
10     manager_id integer,
11     location_id integer,
12     PRIMARY KEY (department_id)
13 );
14
15  CREATE TABLE public.employees
16  (
17     employee_id integer NOT NULL,
18     first_name character varying(20),
19     last_name character varying(25) NOT NULL,
20     email character varying(100) NOT NULL,
21     phone_number character varying(20),
22     hire_date date NOT NULL,
23     job_id character varying(10) NOT NULL,
24     salary numeric(8, 2) NOT NULL,
25     commission_pct numeric(2, 2);
```

Below the script, the 'Messages' tab is active, showing the output: 'COMMIT' and 'Query returned successfully in 99 msec.'

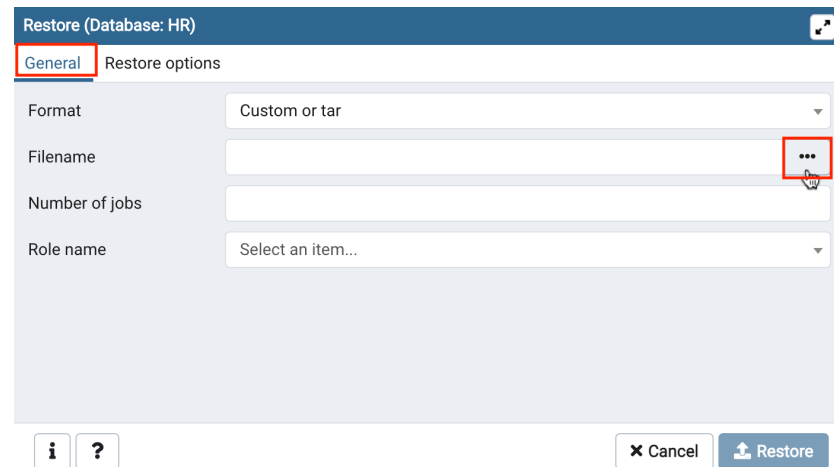
Task C: Load the database schema with data

In this task of the Example Exercise, you will load the database schema you created in Task B of the Example Exercise with data using the pgAdmin Restore feature.

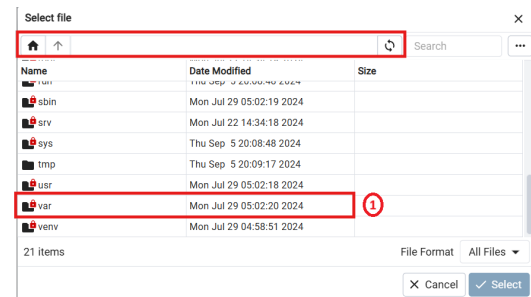
1. Download the [HR_pgsql_dump_data_for_example-exercise.tar](#) PostgreSQL dump file (containing the partial HR database data) using the link below to your local computer.
 - [HR_pgsql_dump_data_for_example-exercise.tar](#)
2. Follow the instructions below to import/restore the data:
 - In the tree-view, expand **HR**. Right-click **HR** and click **Restore**.



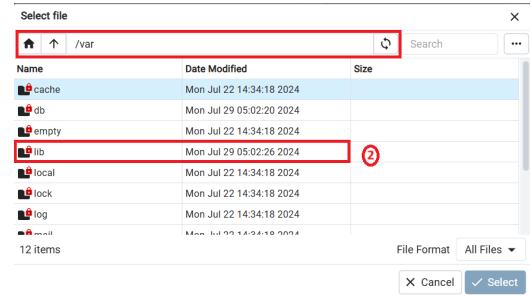
On the General tab, click Select file by the Filename box.



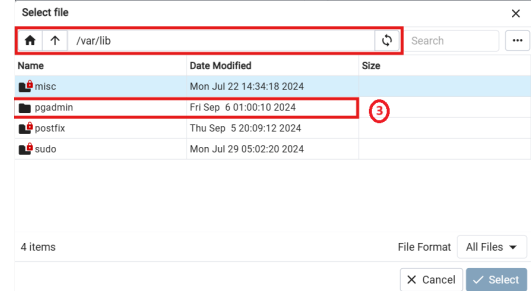
Initially make sure the folder details empty and select the var option from the list as shown in the screenshot below. Select var folder.



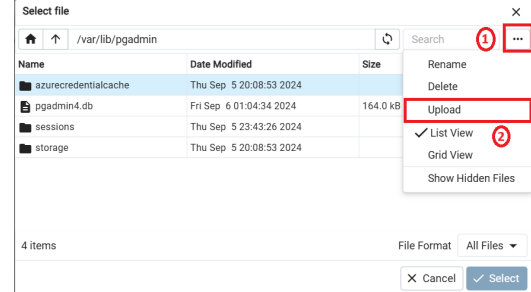
Select lib folder.



Select pgadmin folder. Here you could notice the folders are locked except the pgadmin folder.

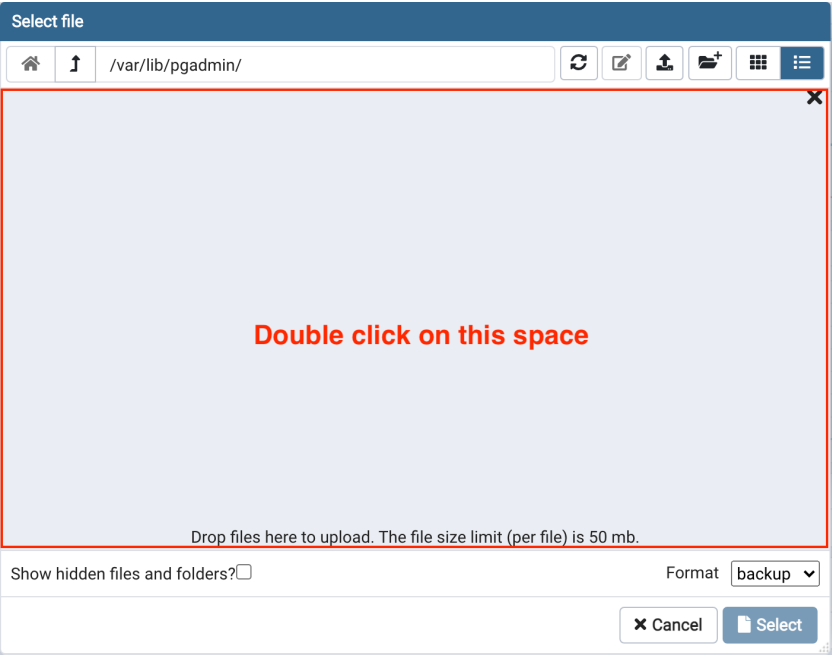


Click Upload File. Now select upload as mentioned here.

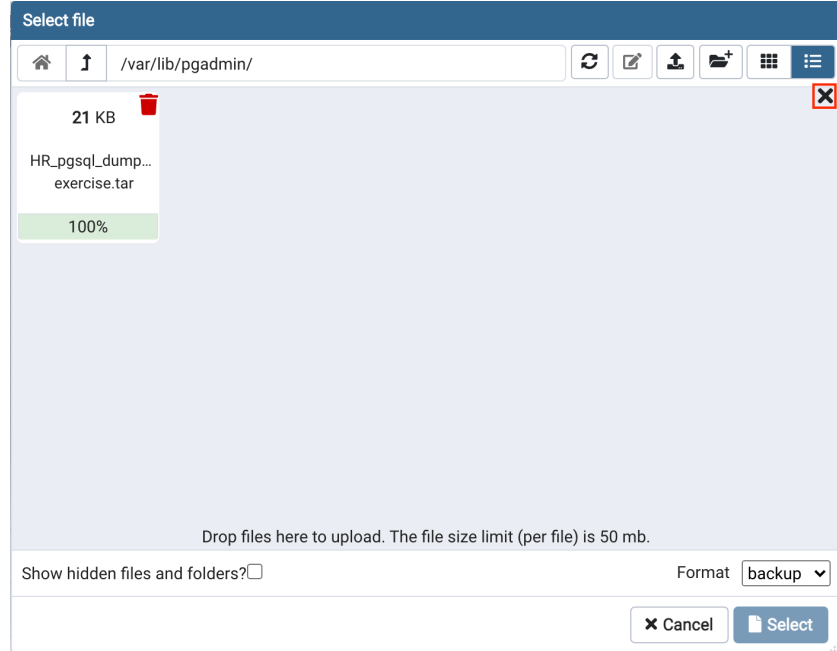


Double-click on the drop files area and load the **HR_pgsql_dump_data_for_example-exercise.tar** you downloaded earlier on your local computer.

Note: Ensure that you upload the files to this path: /var/lib/pgadmin/



◦ When the upload is complete, close the drop files area by clicking X.



◦ Ensure Format is set to All Files, select the uploaded HR_pgsqL_dump_data_for_example-exercise.tar file from the list, and then click Select.

Select file

/var/lib/pgadmin/HR_pgsql_dump_data_for_example-ex...

Name	Size	Modified
 HR_pgsql_dump_data_for_example-exercise.tar	20.5 kB	Thu Apr 1 13:46:45 2021
 pgadmin4.db	156.0 kB	Thu Apr 1 13:45:14 2021
 sessions	4.0 kB	Thu Apr 1 09:25:08 2021
 storage	4.0 kB	Thu Apr 1 09:24:08 2021

Show hidden files and folders?☐

Format

All Files

Cancel

Select

Now switch to the **Restore options** tab.

Restore (Database: HR)

General

Restore options

Format

Custom or tar

Filename

/var/lib/pgadmin/HR_pgsql_dump_data_for_example-exercise.tar

...

Number of jobs

Role name

Select an item...

i

?

Cancel

Restore

Under **Disable**, set the **Trigger** option to **Yes**. Then click **Restore**.

21 of 27

1/2/2026, 11:09 PM

General

Restore options

Queries

Include CREATE DATABASE statement

No

Clean before restore

No

Single transaction

No

Disable Trigger

Yes

No data for Failed Tables

No

i

?

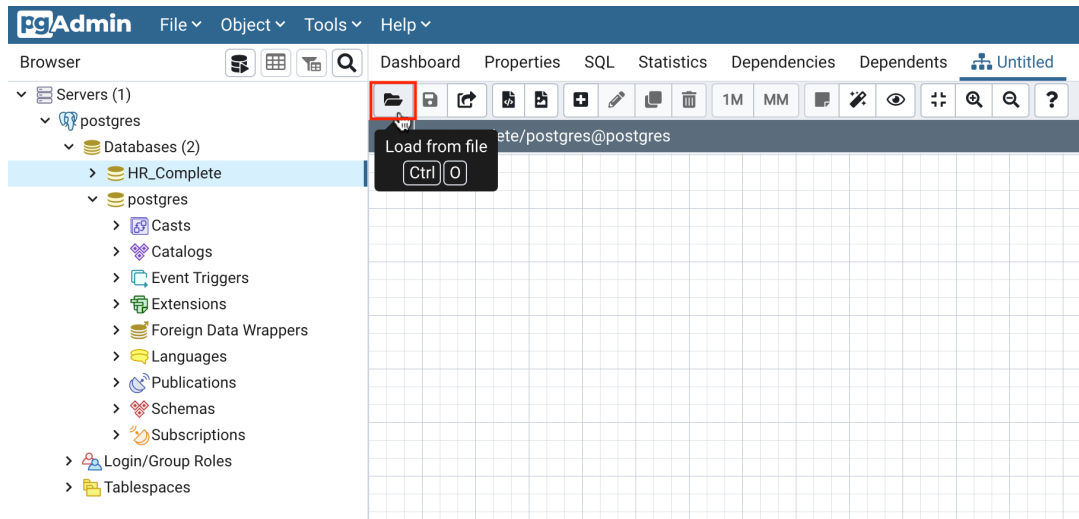
Cancel

Restore

Practice Exercise

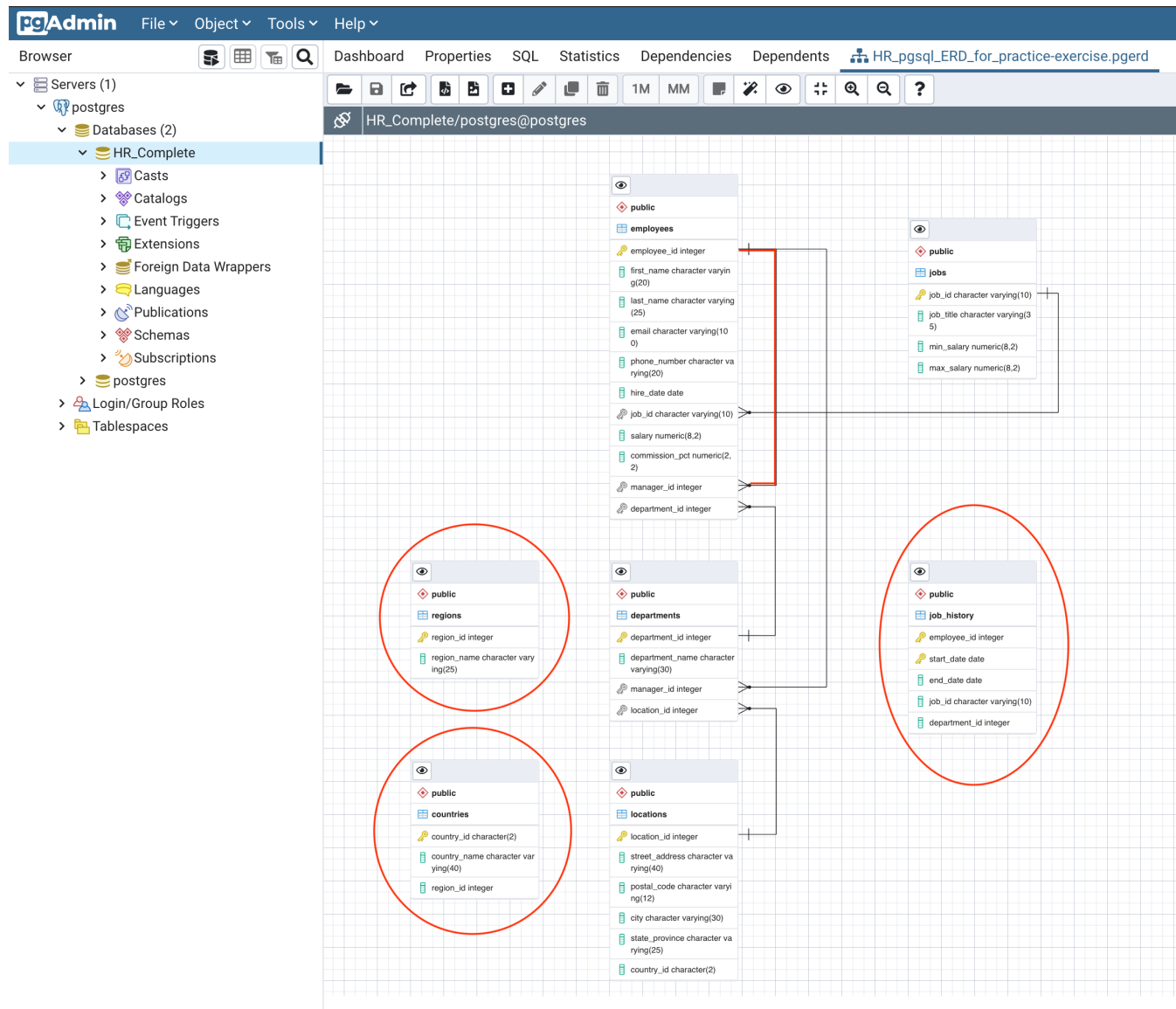
In this practice exercise, first you will finish creating a partially complete ERD for the HR database. Next, you will generate and execute an SQL script to build the complete schema of the HR database from its ERD. Finally, you will load the complete database schema with data by using the Restore feature.

- Download the [HR_pgsql_ERD_for_practice-exercise.pgerrd](#) ERD file (containing a partial HR database ERD based on the one that you created in Task A of the Example Exercise) below to your local computer.
 - [HR_pgsql_ERD_for_practice-exercise.pgerrd](#)
- In pgAdmin, create a new database named **HR_Complete**.
- Open the ERD Tool and use **Load from file** to load the [HR_pgsql_ERD_for_practice-exercise.pgerrd](#) file.



Tip: Follow Example Exercise Task C for how to load any file in pgAdmin.

- You will see the previous four entity diagrams along with relationships that you created in the Example Exercise. You will also see three new entity diagrams for the **job_history**, **regions**, and **countries** tables and one new relationship within the entity diagram of the **employees** table between *manager_id* as local column and *employee_id* as referenced column.



5. Create the remaining relationships between the tables:

- [Click here] Create a relationship between countries and regions

Select the entity diagram **countries** and click **One-to-Many** link. Now enter the definition information for a foreign key on the **countries** table as shown in the image below to create the relationship. Then click **OK**.

One to many relation

General

Local Table

(public) countries

Local Column

region_id

x

Referenced Table

(public) regions

x

Referenced Column

region_id

x

Cancel

OK

► [Click here] Create a relationship between job_history and departments

Select the entity diagram **job_history** and click the **One-to-Many link** button. Now enter the definition information for a foreign key on the **job_history** table as shown in the image below to create the relationship. Then click **OK**.

One to many relation

General

Local Table

(public) job_history

Local Column

department_id

x

Referenced Table

(public) departments

x

Referenced Column

department_id

x

Cancel

OK

► [Click here] Create a relationship between job_history and employees

Select the entity diagram **job_history** and click **One-to-Many link**. Now enter the definition information for a foreign key on the **job_history** table as shown in the image below to create the relationship. Then click **OK**.

One to many relation

General

Local Table

(public) job_history

Local Column

employee_id

x

Referenced Table

(public) employees

x

Referenced Column

employee_id

x

Cancel

OK

► [Click here] Create a relationship between job_history and jobs

Select the entity diagram **job_history** and click **One-to-Many link**. Now enter the definition information for a foreign key on the **job_history** table as shown in the image below to create the relationship. Then click **OK**.

One to many relation

General

Local Table

(public) job_history

Local Column

job_id

x

Referenced Table

(public) jobs

x

Referenced Column

job_id

x

Cancel

OK

► [Click here] Create a relationship between locations and countries

Select the entity diagram **locations** and click **One-to-Many link**. Now enter the definition information for a foreign key on the **locations** table as shown in the image below to create the relationship. Then click **OK**.

One to many relation

General

Local Table

(public) locations

Local Column

country_id

x

Referenced Table

(public) countries

x

Referenced Column

country_id

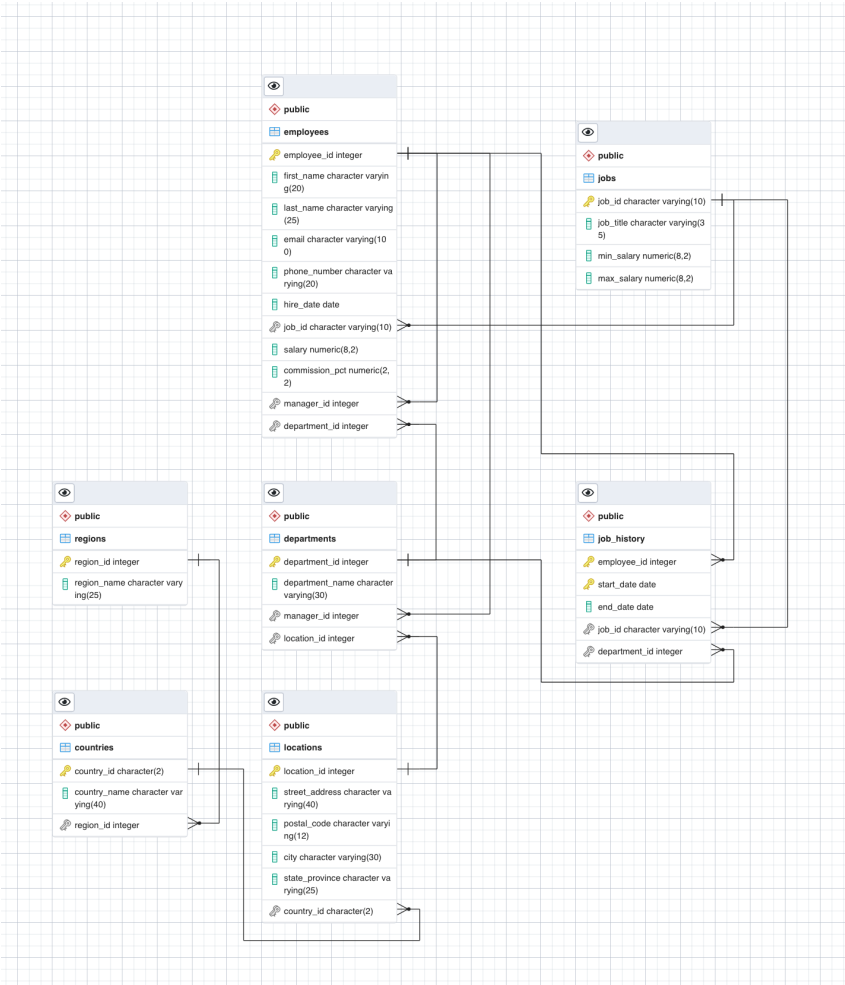
x

✕ Cancel

OK

Tip: Follow Example Exercise Task A for how to create relationships between the entities by adding foreign keys to the tables.

6. After creating the remaining relationships, the complete ERD of the HR database will look like the following image:



7. Generate and execute an SQL script from the ERD to create the schema of the **HR_Complete** database.

Tip: Follow Example Exercise Task B.

8. Download the **HR_pgsql_dump_data.tar** PostgreSQL dump file (containing the complete HR database data) below to your local computer. Use the dump file to restore/import the data to the **HR_Complete** database.

◦ [HR_pgsql_dump_data.tar](#)

Tip: Follow Example Exercise Task C.

Conclusion

Congratulations! You have completed this lab, and you have learned how to create an ERD of a database, generate and execute an SQL script from an ERD to create a schema, and load the database schema with data.

Author(s)

- [Sandip Saha Jey](#)

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